ROS 2 Sensor Run Guide: Terminals 1 to 6

ICM-20948 IMU (I2C bus 2, address 0x68), heading Kalman filter, and Waveshare LG290P GNSS/RTK on the LattePanda 3 Delta. ROS 2 Humble, package **imu_processing**.

Overview

Terminal	Runs	Reads	Publishes / prints
1	imu_raw_live	IMU over I2C	/imu/gyr_raw, /imu/mag_raw
2	imu_cal_live	Terminal 1 topics	/imu/gyr_cal, /imu/mag_cal
3	heading_live	Terminal 2 topics	/imu/heading/gyro, /imu/heading/mag, /imu/heading_kalman
4	print_headings	Terminal 3 topics	Prints + plots compass headings (0-360 deg)
5	nmea_serial_driver	LG290P over USB serial	/gps/fix
6	gps_live	Terminal 5 topic	Prints position every 2 s + plots path

Start terminals **in order** and leave earlier ones running: each one depends on the one before it. Terminals 5 and 6 (GPS) are independent of 1 to 4 and can be run on their own.

Step 0 — in EVERY new terminal

A new terminal does not know about ROS or this workspace until you source them. Run this first in each of the six terminals:

TYPE

```
source /opt/ros/humble/setup.bash && source ~/jane_ros2_ws/install/setup.bash
```

After changing any code, rebuild once and re-source (then restart the changed node):

TYPE

```
cd ~/jane_ros2_ws && colcon build --packages-select imu_processing && source install/setup.bash
```

Terminal 1 — Raw IMU (imu_raw_live)

Reads the ICM-20948 accelerometer, gyroscope and magnetometer and publishes the raw values. Keep this running the whole time - every IMU step below depends on it.

TYPE

```
ros2 run imu_processing imu_raw_live
```

YOU SHOULD SEE

```
[INFO] [imu_raw_live]: ICM-20948 IMU publisher started
[INFO] [imu_raw_live]: Acceleration XYZ Raw: -1.183, 0.048, 9.926 m/s^2
[INFO] [imu_raw_live]: Angular Velocity XYZ Raw: 0.002, 0.009, 0.002 rad/s
[INFO] [imu_raw_live]: Magnetic Field XYZ Raw: 19.35, -9.15, -26.25 uT
```

The three lines repeat every 50 readings (about every 2.5 s). Z acceleration of about 9.8 m/s^2 is gravity and is correct when the IMU is flat.

Do not add `-p enable_plot:=true` here except for a quick look: plotting inside this node slows publishing to a few readings per second and breaks calibration recording.

OSError: [Errno 121] Remote I/O error means the IMU is not answering on I2C - check wiring and 3.3 V power, then run `i2cdetect -y -r 2` and look for 68.

Terminal 2 — Calibrated IMU (imu_cal_live)

Subtracts the gyro bias and applies the magnetometer hard/soft-iron calibration. Needs ~/imu_calibration/bias_file_gyro.csv and ~/imu_calibration/mag_calibration.csv (see One-time calibration on the last page).

TYPE

```
ros2 run imu_processing imu_cal_live
```

YOU SHOULD SEE

```
[INFO] [imu_cal]: Publishing calibrated gyro on /imu/gyr_cal and calibrated
magnetometer data on /imu/mag_cal. Accelerometer values remain unchanged.
```

Prints only this one line, then publishes silently. That is normal.

Optional: to see calibrated IMU plots, open an extra terminal and run `ros2 run imu_processing imu_plot_live` (add `--ros-args -p source:=raw` for raw data).

Terminal 3 — Heading calculation (heading_live)

Integrates the calibrated gyro, computes the magnetometer heading, and fuses both in a Kalman filter.

TYPE

```
ros2 run imu_processing heading_live
```

YOU SHOULD SEE

```
[INFO] [heading_live]: Calculating headings from /imu/gyr_cal and /imu/mag_cal;
publishing radians on /imu/heading/gyro, /imu/heading/mag, and /imu/heading_kalman.
```

Prints only this line. Headings are published in radians, counter-clockwise (ROS convention); Terminal 4 converts them to compass degrees.

Tuning (optional): `--ros-args -p mag_noise:=0.03 -p bias_noise:=0.003 -p angle_noise:=0.001`. Lower mag_noise = follows the magnetometer faster but noisier.

Terminal 4 — Heading display (print_headings)

Prints and plots the three headings as compass bearings: **0 to 360 degrees, clockwise from north**.

TYPE

```
ros2 run imu_processing print_headings

# after north calibration (example value):
ros2 run imu_processing print_headings --ros-args -p north_offset_deg:=37.5
```

YOU SHOULD SEE

```
gyro_head: 315.9 deg, mag_head: 315.5 deg, kalman_head: 315.5 deg
gyro_head: 311.7 deg, mag_head: 311.4 deg, kalman_head: 311.4 deg
```

Prints 5 times a second and opens the **Live IMU Headings** plot (last 30 s; x-axis is seconds ago).

Gyro = smooth but drifts. Magnetometer = absolute but noisy. Kalman = fused best estimate.

Options: `-p enable_plot:=false` (text only), `-p plot_window_s:=60.0`, `-p clockwise_positive:=true` (if turning clockwise makes the number go down).

Terminal 5 — GPS driver (nmea_serial_driver)

Reads NMEA sentences from the LG290P over USB serial (460800 baud) and publishes /gps/fix.

TYPE

```
ros2 run nmea_navsat_driver nmea_serial_driver --ros-args \
    -p port:=/dev/serial/by-id/usb-1a86_USB_Single_Serial_5B91033589-if00 \
    -p baud:=460800 -r fix:=/gps/fix
```

YOU SHOULD SEE

```
[INFO] [nmea_navsat_driver]: Successfully connected to /dev/serial/by-id/... at 460800.
```

Prints no positions itself - that is Terminal 6's job. One 'utf-8 codec can't decode' warning at startup is a half-received first sentence and can be ignored.

The by-id path is used because /dev/ttyACM0 and /dev/ttyACM1 can swap with the Arduino after a reboot.

Could not open port / device busy: another driver is already running. Stop it with `pkill -f nmea_serial_driver` and start again.

Terminal 6 — GPS position (gps_live)

Prints the latest position every 2 seconds and plots the path in metres from the first fix.

TYPE

```
ros2 run imu_processing gps_live

# text only, no plot window:
ros2 run imu_processing gps_live --ros-args -p enable_plot:=false
```

YOU SHOULD SEE

```
lat: -37.79459464 deg, lon: 144.79711584 deg, alt: 41.05 m | east: -0.783 m,
north: -0.596 m | fix: GPS, accuracy: 12.162 m
```

waiting for /gps/fix = Terminal 5 is not running. **NO FIX - waiting for satellites** = driver works but the antenna needs open sky.

Change the print interval with `-p print_period_s:=5.0`.

GPS output units

Field	Unit	Meaning
lat, lon	degrees (WGS84)	Negative latitude = south, positive longitude = east
alt	metres	Height above the WGS84 ellipsoid (not above sea level)
east, north	metres	Distance from the first valid fix after gps_live started
fix	-	NO FIX / GPS / DGPS/SBAS / RTK (RTK needs a correction stream)
accuracy	metres	Estimated horizontal error (1 sigma) from the driver - an estimate, not measured

Without RTK corrections the LG290P gives a standard GPS fix (about 2 to 15 m). Centimetre accuracy (fix: RTK, accuracy about 0.02 m) requires RTCM correction data from a base station or NTRIP service.

IMU and heading units

Value	Printed unit	Topic unit
Acceleration	m/s^2	m/s^2
Angular velocity	rad/s	rad/s
Magnetic field	uT (microtesla)	T (tesla)
Headings	degrees, 0-360, clockwise from N	radians, -pi to pi, counter-clockwise

One-time calibration

Gyro bias (IMU completely still, Terminal 1 running)

TYPE

```
ros2 run imu_processing cal_gyro_record
```

Keep the IMU completely still for 60 to 90 s, then press Ctrl+C. It needs at least 1000 samples (about 50 s at the current ~21 Hz) or the bias file is NOT updated. Saves ~/imu_calibration/bias_file_gyro.csv. Restart Terminal 2 afterwards.

Magnetometer (Terminal 1 running, no plots)

1. Start the recorder. 2. Press **s**, slowly rotate the IMU through every orientation for about 60 s, away from metal. 3. Press **s** again (expect hundreds to thousands of rows, not ~30). 4. Press **q**. 5. Run the fit on the newest recording.

TYPE

```
ros2 run imu_processing cal_mag_record
ros2 run imu_processing calibrate_mag "$(ls -t ~/mag_recordings/mag_*.csv | head -1)"
```

Saves ~/imu_calibration/mag_calibration.csv. Restart Terminal 2 afterwards.

North offset for Terminal 4

1. Point the rover's front at north using a phone compass (set it to true north for true bearings). 2. Read the **mag_head** value printed in Terminal 4. 3. Restart Terminal 4 with `-p north_offset_deg:=<that value>`. 4. Turn the rover clockwise toward east: the number should go **up**. If it goes down, add `-p clockwise_positive:=true` and repeat step 2.

Stopping and troubleshooting

Problem	Fix
Stop a node	Press Ctrl+C in its terminal, or close its plot window.
Plot window frozen or black	pkill -f print_headings (or gps_live / imu_plot_live), then run it again.
'Package not found' or old behaviour	Run the Step 0 source line again; rebuild if code changed.
bash: BUS / recording: No such file	You copied a placeholder like <BUS>. Type a real value, without < >.
No headings printed	Check Terminals 1, 2 and 3 are all running.
Check a topic directly	ros2 topic echo /gps/fix or ros2 topic hz /imu/gyr_raw